

FIITJEE

ALL INDIA TEST SERIES

FULL TEST – V

JEE (Main)-2024

TEST DATE: 17-01-2024

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1. C

Sol. $v = v_c - \omega R = 0$

In the frame of centre of mass, which is moving with constant velocity V_c .

Centripetal acceleration = $\frac{V_c^2}{R}$ upward.

$\therefore$ Required acceleration = $\frac{V_c^2}{R}$ upward.

2. A

Sol. Energy density = $\frac{1}{2} \epsilon_0 E_o^2$ and $E_o = (E_{rms})\sqrt{2}$

3. C

Sol. From conservation of angular momentum

$$I\omega = I_o\omega_o$$

$$\therefore \frac{1}{2} I \omega^2 = \frac{1}{2} \times I \frac{I_o^2 \omega_o^2}{I^2} = \frac{1}{2} I_o \omega_o^2 \cdot \left(\frac{I_o}{I} \right)$$

$$(K.E.)_{final} < (K.E.)_{initial} ; I > I_o$$

$$\therefore W_{man} = -ve.$$

4. D

Sol. Just before striking velocity of bob

$$V = \sqrt{2gh}$$

$$\text{By COM } mv = (m + m)v'$$

$$\text{By COME } \frac{1}{2}(2m)(v')^2 = (2m)gH$$

5. C

Sol. For the system of two particles gravitational energy is given as

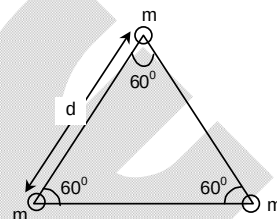
$$U = -Gm_1m_2/r$$

$$U_A = U_2 + U_{23} + U_{31}$$

$$U_A = -3 \frac{GMm}{d} = -\frac{3GM^2}{d}$$

The answer is (C).

-ve sign resembles that the particles are bounded by their mutual gravitational field.



6. B

Sol. 10 kg mass will gain velocity along the string only. Using momentum conservation along the string

$$v = \frac{10 \cos 60^\circ}{10 + 5} = \frac{1}{3} \text{ m/s}$$

7. D

$$\text{Sol. } \frac{mv^2}{(R+x)} = \frac{gR^2m}{(R+x)^2} \Rightarrow V^2 = \frac{gR^2}{(R+x)} \Rightarrow V = \left(\frac{gR^2}{R+x} \right)^{\frac{1}{2}}$$

8. B

$$\text{Sol. } T = 2\pi \sqrt{\frac{m}{k}}$$

$$3 = 2\pi \sqrt{\frac{0.9}{k}}$$

$$T = 2\pi \sqrt{\frac{0.4}{k}}$$

$$\Rightarrow \frac{T}{3} = \sqrt{\frac{0.4}{0.9}} = \frac{2}{3} \Rightarrow T = 2 \text{ sec}$$

9. A

Sol. As shown in the figure, angle of incidence will be minimum for the ray travelling along the inner edge. So the whole incident light entering the glass through surface A will emerge from the glass through surface B.

$$\theta > i_c$$

$$\sin \theta > \sin i_c$$

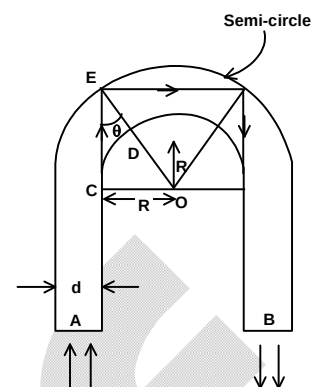
$$\text{But as, } \sin i_c = \frac{1}{\mu} = \frac{2}{3}$$

And from the geometry of the figure,

$$\sin \theta = \frac{OC}{OE} = \frac{R}{OD + DE} = \frac{R}{R + d}$$

$$\text{so, } \frac{R}{R + d} > \frac{2}{3}, \text{ i.e. } R > 2d$$

$$\text{i.e., } \frac{d}{R} < \frac{1}{2} \text{ so } \left(\frac{d}{R} \right) = \frac{1}{2}$$



10. C

Sol. For charging $q = CE(1 - e^{-t/RC})$

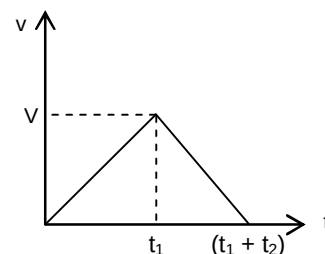
Charge at $t = RC \Rightarrow q_0 = CE(1 - e^{-1})$

At $t = RC$ discharging starts

$$\Rightarrow q = q_0(e^{-t/RC}) = CE(1 - e^{-1}) \times \frac{1}{e} = CE \left(\frac{1}{e} - \frac{1}{e^2} \right)$$

11. A

Sol. Required ratio = $\frac{V}{\text{Area} / (t_1 + t_2)} = \frac{2}{1}$



12. B

Sol. Antinodes are formed at a distance $0.6r$ from the end of the tube, where r is radius of the tube.

$$47 + 2 \times 0.6r = \lambda/2$$

$$\lambda = v/n = 100 \text{ cm}$$

$$47 + 0.6D = 100/2$$

$$D = 5 \text{ cm}$$

13. B

Sol. $dV = -IdR = -I\rho \frac{dx}{A}$

$$\frac{dV}{dx} = -I \frac{\rho}{A} = -k$$

$$\Rightarrow R = k\ell/l$$

II Method:

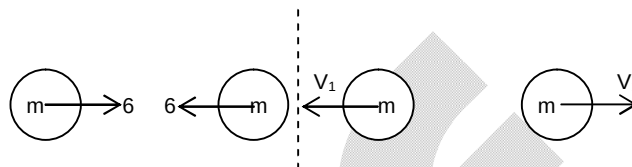
$$V_2 - V_1 = \int \left(\frac{dV}{d\ell} \right) d\ell = kL = IR$$

14. B

$$\text{Sol. } V_1 = \left(\frac{m_1 - em_2}{m_1 + m_2} \right) u_1 + \left(\frac{m_2 + em_2}{m_1 + m_2} \right) u_2$$

$$\left(\frac{1-e}{2} \right) \times 6 + \left(\frac{1+e}{2} \right) (-6)$$

$$= \frac{2}{6} \times 6 + \frac{4}{6} \times (-6) = -2 \text{ m/s}$$



15. A

$$\text{Sol. Total resistance} = \frac{500}{3} + \frac{\left(100 + \frac{100^2}{25} \right) \left(600 + \frac{200^2}{100} \right)}{100 + \frac{100^2}{25} + 600 + \frac{200^2}{100}} = 500$$

$$\text{Required heat} = \frac{200^2}{500} = 80 \text{ J.}$$

16. C

$$\text{Sol. For closed organ pipe } f_c = (2n-1) \frac{v}{4L_c} \text{ with } n = 1, 2, \dots$$

$$\text{While for open organ pipe } f_o = n \frac{v}{2L_o} \text{ with } n = 1, 2, \dots$$

According to given problem

$$f_c = \frac{v}{4L_c} = 110 \quad \text{i.e.,} \quad L_c = \frac{330}{4 \times 110} = 0.75 \text{ m}$$

Furthermore

$$\frac{3v}{4L_c} \sim \frac{2v}{2L_o} = 2.2 \quad \text{or} \quad \frac{330}{L_o} = 330 \mp 2.2 \quad \left(\text{as } \frac{3v}{4L_c} = 3f_c = 330 \right)$$

$$\text{i.e., } L_o = \frac{330}{330 \mp 2.2} \Rightarrow L_o = 1.00067 \text{ m} \quad \text{or} \quad 0.9937 \text{ m}$$

17. A

$$\text{Sol. } R = \frac{V^2}{gV_o} \cdot V = \frac{V^3}{gV_o} ; R \propto V^3$$

18. A

$$\text{Sol. } p_o + 2\rho gh - 2\rho a_o l/2 - \rho a_o l/2 - \rho gh = p_o$$

$$h = \frac{3a_o l}{2g}$$

19. C

Sol. Potential energy of particle at the centre of earth is $U = -\frac{3}{2} \frac{GMm}{R_e}$

$$\text{So, } V_e = \sqrt{\frac{3GM}{R_e}} = \sqrt{3gR_e}$$

20. A

Sol. The equivalent capacitance

$$C = \frac{4\pi\epsilon_0(3a \times 4a)}{(4a - 3a)} + 4\pi\epsilon_0(4a)$$

$$= 64\pi\epsilon_0 a$$

SECTION - B

21. 4

Sol. $V = L \left(\frac{di}{dt} \right)$

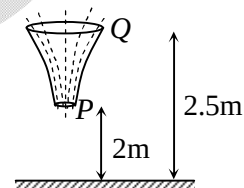
22. 1

Sol. $P_1 - P_2 = \frac{1}{2} \rho (v_2^2 - v_1^2) + \rho g (h_2 - h_1)$

$$= \frac{1}{2} \times 10^3 (1^2 - 3^2) + 10^3 \times 10 (0.5)$$

$$= -4 \times 10^3 + 5 \times 10^3$$

$$= 1000 \text{ J/m}^3$$



23. 2

Sol. $\int \tau dt = NAB \int i dt$

$$I\omega_0 = NABQ \quad \dots\dots(i)$$

$$\frac{1}{2} I\omega_0^2 = \frac{1}{2} C\theta_{\max}^2 \quad \dots\dots(ii)$$

From equation (i) and (ii)

$$\theta_{\max} = \frac{NABQ}{\sqrt{CI}}$$

24. 2

Sol. $F = \frac{1}{4\pi\epsilon_0} \frac{q(Q-q)}{r^2}$

For maximum repulsion force $\frac{dF}{dq} = Q - 2q = 0$

$$\frac{Q}{q} = \frac{2}{1} = 2$$

25. 3

Sol. $i = \frac{2I_0}{T} t \quad [0 < t < T/2]$

$$i_{rms} = \frac{\frac{2I_0}{T} \int_0^{T/2} t^2 dt}{\sqrt{T/2}} = \frac{I_0}{\sqrt{3}}.$$

26. 4

Sol. The charge q is completely closed by four identical given box. Therefore, the required flux = $\frac{1}{4} \frac{q}{\epsilon_0}$.

27. 2

Sol. $F = i2\sqrt{2} B = 10 \times 2\sqrt{2} \times 10$
 $= 200\sqrt{2} = 2 \times 100\sqrt{2}$

28. 5

Sol. At stable equilibrium, U is minimum.

$$\frac{dU}{dx} = 0 \text{ and } \frac{d^2U}{dx^2} > 0$$

$$= \frac{1}{dx} \left(\frac{x^3}{3} - \frac{ax^2}{2} + 20x \right) = 0.$$

$$\Rightarrow x^2 - 9x + 20 = 0. \Rightarrow (x-5)(x-4) = 0.$$

$x = 5$ and $x = 4$ are points of equilibrium.

And U minimum when $\frac{d^2U}{dx^2} > 0$. i.e. at $x = 5$.

29. 1

Sol. $V_A - V_B = 6V$ for current in 4Ω resistor to be zero.

Potential difference 2Ω resistor = $10 - 6 = 4 V$

Current through 2Ω resistor = $2 A$

$$\text{So, } 2 = \frac{\text{net emf}}{\text{net resistance}} = \frac{(10-4)}{(2+R)}$$

On solving, $R = 1 \Omega$

30. 3

Sol. $2000 \times 15 \times 0.5 + m \times 80 = 2500 \times 25 \times 1$

$$m = 500 \text{ gm}$$

$$\text{Water} = 2.5 \text{ kg} + 0.5 \text{ kg} = 3 \text{ kg}.$$

Chemistry**PART – B****SECTION – A**

31. B

Sol. $r_n = x \times \frac{n^2}{Z}$
 $= x \times \frac{(1)^2}{3} = \left(\frac{x}{3}\right) \text{Å}$

32. B

Sol. The unit of k will be $\text{mol}^{-1/2} \text{L}^{1/2} \text{s}^{-1}$.

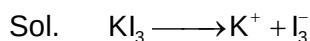
33. B

Sol. It is a zero order reaction.

34. B

Sol. The N atom in NO_2^+ undergoes sp-hybridization with bond angle 180° .

35. C



$\therefore$ The oxidation number of iodine in I_3^- is $-\frac{1}{3}$

36. D

Sol. K_a increases only by increasing temperature.

37. C

Sol. Milliequivalent of $\text{H}_2\text{SO}_4 = 25.1 \times 0.02 = 0.502$

Equivalent = 0.502×10^{-3}

Eq of $\text{H}_2\text{SO}_4 = \text{Eq of CaCO}_3$

$\therefore$ Mass of $\text{CaCO}_3 = \text{Eq} \times 50 = 0.502 \times 10^{-3} \times 50 = 0.0251$

100 mL $\text{H}_2\text{O} = 100 \text{ g}$ contains 0.0251 g CaCO_3

$10^6 \text{ g of water will contain } \frac{0.0251}{10^2} \times 10^6 = 251 \text{ g of CaCO}_3$

 $\therefore$ The degree of hardness = 251 ppm

38. C

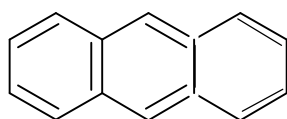
Sol. P_4O_{10} removes water from HNO_3 forming N_2O_5 .

39. B

Sol. It produces nascent oxygen (O) which can bleaching colouring matter.

40. D

Sol.



41. D

 Sol. The products are due to coupling of different alkyl groups. If $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}$ is RCl and $\text{CH}_3\text{CH}(\text{Cl})\text{CH}_3$ is $\text{R}'\text{Cl}$, the products will be RR , $\text{R}'\text{R}'$ and RR' .


42. C

 Sol. NH_3 does not have suitable orbital to form backbond.

43. B

Sol. Due to intermolecular hydrogen bonding.

44. A

Sol. Aliphatic amines are stronger base than aromatic amines

45. C

 Sol. $\text{CH}_3\text{CHO} \xrightarrow[\text{KCN}]{\text{NH}_4\text{Cl}} \text{MeCH}(\text{NH}_2)\text{CN} \xrightarrow[\text{H}_2\text{O}]{\text{HCl}} \text{MeCH}(\text{NH}_2)\text{COOH}$

46. B

 Sol. $4\text{BCl}_3 + 6\text{H}_2 + \text{C} \longrightarrow \text{B}_4\text{C} + 12\text{HCl}$

 The reaction takes place in the preparation of bullet proof fabric from B_4C .

47. D

 Sol. This is due to lanthanoid contraction on 6th period.

48. B

Sol. The configuration of carbon atoms at 3, 4, 5 positions should be same.

49. C

Sol. Tertiary alcohols do not undergo oxidation.

50. C

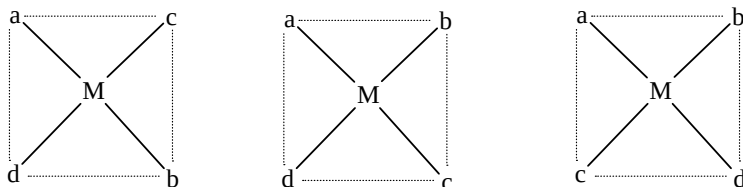
 Sol. The functional isomer of $\text{C}_2\text{H}_5\text{NH}_2$ is CH_3NHCH_3 which is more basic than $\text{C}_2\text{H}_5\text{NH}_2$.

SECTION – B

51. 3

Sol. $[\text{Pt}(\text{NH}_3)(\text{NH}_2\text{OH})(\text{NO}_2)(\text{py})]\text{NO}_2$

All four ligands are different, so it can be compared to M abcd. Three geometrical isomers are possible as follows



52. 3

Sol. $E_{\text{Cell}} = E_{\text{Cell}}^{\circ} - \frac{0.0591}{n} \log \frac{[\text{Zn}^{2+}]}{[\text{H}^+]^2}$

$$\text{or, } 0.6418 = [0 - (-0.76)] - \frac{0.0591}{2} \log \frac{10^{-2}}{[\text{H}^+]^2}$$

on solving, $[\text{H}^+]^2 = 10^{-6}$

$\therefore [\text{H}^+] = 10^{-3}$, pH = 3

53. 724

Sol. The elevation in boiling point is

$$\Delta T_b = K_b m : m = \text{molality} \frac{n_2}{w_1} \times 1000$$

$$2 = 0.76 \times \frac{n_2}{100} \times 1000$$

$$n_2 = \frac{5}{19}$$

Also, from Raoult's law of lowering of vapour pressure

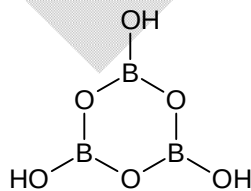
$$\frac{-\Delta p}{p^{\circ}} = x_2 = \frac{n_2}{n_1 + n_2} = \frac{n_2}{n_1}$$

$$\Rightarrow -\Delta p = 760 \times \frac{5}{19} \times \frac{18}{100} = 36 \text{ mm of Hg}$$

$$\Rightarrow p = 760 - 36 = 724 \text{ mm of Hg}$$

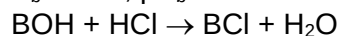
54. 132

Sol. Structure of metaboric acid is



Molar mass = 132

55. 9

 Sol. $K_b = 10^{-5}$, $pK_b = 5$


At half neutralization, 50% of the base is converted to its salt, with strong acid HCl, it forms a basic buffer.

$$pOH = pK_b + \log \frac{[\text{salt}]}{[\text{base}]}$$

$$pOH = 5 + \log \frac{[\text{salt}]}{[\text{base}]}$$

$$[\text{salt}] = [\text{base}]$$

$$pOH = 5$$

$$pH = 14 - pOH = 9$$

56. 3

 Sol. $n = 3$, $\ell = 1$ and $m = -1$, for the lowest or minimum sum.

57. 352

 Sol. $\Delta H = 70.4 \text{ kJ mol}^{-1} = 70400 \text{ J mol}^{-1}$

$$\Delta S = 2(400) - (250 + 350) = 200 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$T = \frac{\Delta H}{\Delta S} \text{ (as } \Delta G = \Delta H - T\Delta S = 0 \text{ at equilibrium)} = \frac{70400}{200} = 352 \text{ K}$$

58. 195

 Sol. Formula is $\text{Be}_3\text{Si}_2\text{O}_7$

59. 20

Sol. Half-life of first order reaction does not depend on concentration of reactant.

60. 10

 Sol. $\text{NaCN} + \text{HCl} \longrightarrow \text{NaCl} + \text{HCN}$

$$\begin{array}{cccc} 1 \times 110 & V \times 1 & 0 & 0 \\ 110 - V & 0 & V & V \end{array}$$

$$pH = pK_a + \log \frac{[\text{Salt}]}{[\text{Acid}]} = pK_a + \log \frac{110 - V}{V}$$

$$\text{For highest pH, } \frac{[\text{Salt}]}{[\text{Acid}]} = \frac{10}{1} = \frac{110 - V}{V}$$

$\therefore$ On solving, $V = 10 \text{ mL}$

Mathematics**PART – C****SECTION – A**

61. D

Sol. Put $x = r \cos \theta, y = r \sin \theta$

$$\Rightarrow 2013r^2 - 2004r^2 \sin 2\theta = 1$$

$$\Rightarrow r^2 = \frac{1}{2013 - 2004 \sin 2\theta}$$

r is maximum, when $2013 - 2004 \sin 2\theta$ is minimum.

$$2013 - 2004 \sin 2\theta \big|_{\min} = 9 \Rightarrow r^2 \big|_{\max} = \frac{1}{9} \Rightarrow r \big|_{\max} = \frac{1}{3}$$

62. C

Sol. $R = \{(1, 3), (2, 5)\}$

Obviously, R is not reflexive or symmetric

R is trivially transitive.

63. A

Sol. Using $(a + b)^7 - (a - b)^7$

$$= 2[{}^7C_1 a^6 b + {}^7C_3 a^4 b^3 + {}^7C_5 a^2 b^5 + {}^7C_7 b^7],$$

$$\text{We get } P(x) = {}^7C_1(x^5 - 1)^3 x + {}^7C_2(x^5 - 1)^2 x^3 + {}^7C_3(x^5 - 1)x^5 + {}^7C_7 x^7$$

Which is a polynomial of degree 16.

64. B

Sol. Let the numbers selected by x_1, x_2, x_3

$$\text{We must have } 2x_2 = x_1 + x_3$$

$$\Rightarrow x_1 + x_3 = \text{even}$$

$\Rightarrow x_1, x_3$ both are even, we can select them in ${}^{12}C_2$ ways. Similarly, if x_1 and x_2 both are odd, we can again select them in ${}^{12}C_2$ ways.

$$\text{Thus, total ways} = 2 \cdot {}^{12}C_2 = 132$$

65. C

Sol. $[x]$ is not differentiable at integral points. Also $[n + p \sin x] = n + [p \sin x]$

$\therefore [p \sin x]$ is not differentiable, where $p \sin x$ is an integer. But p is prime and

$$0 < \sin x \leq 1 \quad [\because 0 < x < \pi]$$

$\therefore p \sin x$ is an integer only when $\sin x = \frac{r}{p}$, where $0 < r \leq p$ and $r \in \mathbb{N}$

$$\text{For } r = p, \sin x = 1 \Rightarrow x = \frac{\pi}{2} \in (0, \pi)$$

$$\text{For } 0 < r < p, \sin x = \frac{r}{p}$$

$$\therefore x = \sin^{-1} \frac{r}{p} \text{ or } \pi - \sin^{-1} \frac{r}{p}$$

Number of such values of $x = p - 1 + p - 1 = 2p - 2$

$\therefore$ Total number of points, where $f(x)$ is not differentiable are $1 + 2p - 2 = 2p - 1$.

66. C

Sol. $(x - 2)(y - 4) = 4$ is the given hyperbola

$\Rightarrow (x - 2)(y - 4) = 0$ is the equation of pair of asymptotes

$\Rightarrow (x - 2)(y - 4) = -4$ is the equation of the conjugate hyperbola

67. C

Sol. Any curve through the intersection of given curves is

$$(2x + 3y + 19)(9x + 6y - 17) + \lambda xy = 0.$$

For this curve to represent a circle coefficient of $x^2 =$ coefficient of y^2 and coefficient of xy term = 0

$$\Rightarrow 12 + 27 + \lambda = 0 \Rightarrow \lambda = -39$$

$\therefore$ Equation of required circle is

$$(2x + 3y + 19)(9x + 6y - 17) - 39xy = 0$$

$$\Rightarrow 18x^2 + 12xy - 34x + 27xy + 18y^2 - 51y + 171x + 114y - 323 - 39xy = 0$$

$$\Rightarrow 18x^2 + 18y^2 + 137x + 63y - 323 = 0$$

68. A

Sol. $4^x - (a - 3)2^x + (a - 4) = 0$

$$x \leq 0, \text{ Let } y = 2^x$$

$$y^2 - (a - 3)y + (a - 4) = 0$$

The roots of Quadratic must lie b/w (0, 1]

(A) $(a - 3)^2 - 4(a - 4) \geq 0$

$$a^2 + 9 - 6a - 4a + 16 \geq 0$$

$$a^2 - 10a + 25 \geq 0$$

$$a \in \mathbb{R}$$

(B) $0 < \frac{a - 3}{2} \leq 1$

$$0 < a - 3 \leq 2$$

$$0 < a \leq 5$$

(C) $f(0) = a - 4 > 0 \quad a > 4$

(D) $f(1) = 1 - a + 3 + a - 4 \geq 0 \quad a \in (4, 5]$

69. D

Sol. $a + b + c \leq 20$

$$\Rightarrow a + b + c + d = 20, a, b, c \geq 1, d \geq 0$$

$$\Rightarrow a^1 + b^1 + c^1 + d = 17, a^1, b^1, c^1, d \geq 0$$

$$\text{No. of solutions} = {}^{17+4-1}C_{4-1} = {}^{20}C_3 = 1140$$

70. B

Sol. Given ${}^nC_2 = 153$

$$\Rightarrow n^2 - n - 306 = 0 \Rightarrow n = 18.$$

71. B

$$\text{Sol. } \cot^{-1} \frac{5}{\sqrt{3}} + \cot^{-1} \frac{9}{\sqrt{3}} + \cot^{-1} \frac{15}{\sqrt{3}} + \cot^{-1} \frac{23}{\sqrt{3}} + \dots \infty$$

$$= \tan^{-1} \left(\frac{\frac{2}{\sqrt{3}} - \frac{1}{\sqrt{3}}}{1 + \frac{2}{\sqrt{3}} \times \frac{1}{\sqrt{3}}} \right) + \tan^{-1} \left(\frac{\frac{3}{\sqrt{3}} - \frac{2}{\sqrt{3}}}{1 + \frac{3}{\sqrt{3}} \times \frac{2}{\sqrt{3}}} \right) + \dots + \dots + \tan^{-1} \left(\frac{\frac{n+1}{\sqrt{3}} - \frac{n}{\sqrt{3}}}{1 + \frac{n+1}{\sqrt{3}} \times \frac{n}{\sqrt{3}}} \right) + \dots$$

$$\Rightarrow S_n = \tan^{-1} \frac{n+1}{\sqrt{3}} - \tan^{-1} \frac{1}{\sqrt{3}}$$

$$\lim_{n \rightarrow \infty} S_n = \frac{\pi}{2} - \frac{\pi}{6} = \frac{\pi}{3}$$

72. B

$$\text{Sol. It is equal to } \int_0^1 \left(\frac{x^3}{4} - x \left(f(x) - \frac{x}{2} \right)^2 \right) dx \text{ which is less than } \int_0^1 \frac{x^3}{4} dx = \frac{1}{16}$$

73. A

$$\text{Sol. } f(x) = a_n x^n + a_{n-1} x^{n-1} + a_{n-2} x^{n-2} + \dots + a_0, [x] = m \text{ then } f(m) = a_n m^n + a_{n-1} m^{n-1} + \dots + a_0$$

$$f(f(m)+1) = a_n (f(m)+1)^n + a_{n-1} (f(m)+1)^{n-1} + \dots + a_0$$

$$kf(m) + a_n + a_{n-1} + \dots + a_0 = kf(m) + f(1)$$

$$\text{Since } \frac{f(f(m)+1)}{f(m)} = k + \frac{f(1)}{f(m)} = \text{integer}$$

$$\Rightarrow m = 1 \text{ hence } x \in [1, 2)$$

74. A

$$\text{Sol. Total such numbers are } (3 \times 2) \times (3!) = 36$$

$$\text{Sum of all such numbers } 66666 \times 18 = 1199988$$

75. D

$$\text{Sol. } f(x) = \frac{1}{2} \ln \left(\sqrt{\sqrt{x^2+1}+x} \pm \sqrt{\sqrt{x^2+1}-x} \right)^2$$

$$= \frac{1}{2} \ln \left\{ \sqrt{x^2+1}+x + \sqrt{x^2+1}-x \pm 2\sqrt{(\sqrt{x^2+1})^2 - x^2} \right\}$$

$$= \frac{1}{2} \ln \{ 2\sqrt{x^2+1} \pm 2 \}$$

An even function hence neither one – one nor onto.

76. C

$$\text{Sol. } I = \int \frac{\cos x + \sqrt{3}}{\left(1 + 2 \sin\left(x + \frac{\pi}{3}\right)\right)^2} dx = \int \frac{\cos x + \sqrt{3}}{(1 + \sin x + \sqrt{3} \cos x)^2} dx$$

$$\begin{aligned} \text{Dividing by } \sin^2 x, \text{ we get } I &= \int \frac{\operatorname{cosec} x \cot x + \sqrt{3} \operatorname{cosec}^2 x}{(\operatorname{cosec} x + 1 + \sqrt{3} \cot x)^2} dx = \frac{1}{\operatorname{cosec} x + 1 + \sqrt{3} \cot x} + c \\ &= \frac{\sin x}{1 + 2 \sin\left(x + \frac{\pi}{3}\right)} + c \end{aligned}$$

77. B

Sol. Odd power of a skew symmetric matrix is always a skew symmetric matrix

78. C

Sol. Plane containing the line will be given as $(4x - 2y - 10) + \lambda(5y - 4z + 3) = 0$

$$\Rightarrow 4x + (5\lambda - 2)y + 4\lambda z + 3\lambda - 10 = 0$$

The plane passing through $(4, 3, 7)$ will give λ

$$\Rightarrow 16 + (5\lambda - 2)y + 4\lambda z + 3\lambda - 10 = 0$$

$$\Rightarrow -10\lambda = 0 \Rightarrow \lambda = 0$$

$$\text{i.e. } 4x - 2y - 10 = 0$$

79. D

$$\text{Sol. Let } y = f^{-1}(8x) \Rightarrow 8x = f(y) = 8y^3 + 3y = 8y^3 \left(1 + \frac{3}{8y^2}\right)$$

$$\Rightarrow 2x^{1/3} = 2y \left(1 + \frac{3}{8y^2}\right)^{1/3} \Rightarrow y = \frac{x^{1/3}}{\left(1 + \frac{3}{8y^2}\right)^{1/3}}$$

$$\text{Similarly, let } z = f^{-1}(x) \Rightarrow x = f(z) = 8z^3 + 3z = 8z^3 \left(1 + \frac{3}{8z^2}\right)$$

$$\Rightarrow x^{1/3} = 2z \left(1 + \frac{3}{8z^2}\right)^{1/3} \Rightarrow z = \frac{x^{1/3}}{2 \left(1 + \frac{3}{8z^2}\right)^{1/3}}$$

$$\text{Thus, } \lim_{x \rightarrow \infty} \frac{f^{-1}(8x) - f^{-1}(x)}{x^{1/3}}$$

$$= \lim_{x \rightarrow \infty} \frac{y - z}{x^{1/3}}$$

$$= \lim_{x \rightarrow \infty} \frac{y - z}{x^{1/3}}$$

$$\begin{aligned}
 &= \lim_{\substack{y \rightarrow \infty \\ z \rightarrow \infty}} \left(\frac{1}{\left(1 + \frac{3}{8y^2}\right)^{1/3}} - \frac{1}{2\left(1 + \frac{3}{8z^2}\right)^{1/3}} \right) \\
 &= 1 - \frac{1}{2} \\
 &= \frac{1}{2}
 \end{aligned}$$

80. A

Sol. $\arg(z_2) + \arg(z_4) - \arg(z_1) - \arg(z_3) = \pi$

$$\Rightarrow \arg\left(\frac{z_2 z_4}{z_1 z_3}\right) = \pi \Rightarrow \frac{z_2 z_4}{z_1 z_3} = -k \text{ (where } k > 0)$$

$$\Rightarrow k = \left| \frac{z_2 z_4}{z_1 z_3} \right| = 1$$

$$\Rightarrow z_2 z_4 = -z_1 z_3$$

$$\begin{aligned}
 &|z_1 z_3 - (z_1 - z_3)(z_2 - z_4)|^2 - |(z_1 - z_3)(z_2 - z_4)|^2 \\
 &= |z_1 z_3|^2 + |(z_1 - z_3)(z_2 - z_4)|^2 - \bar{z}_1 \bar{z}_3 (z_1 - z_3)(z_2 - z_4) \\
 &\quad - z_1 z_3 (\bar{z}_1 - \bar{z}_3)(\bar{z}_2 - \bar{z}_1) - |(z_1 - z_3)(z_2 - z_4)|^2 \\
 &= 1 - \left\{ \frac{(z_1 - z_3)(z_2 - z_4)}{z_1 z_3} + \frac{(z_1 - z_3)(z_2 - z_4)}{z_2 z_4} \right\} = 1
 \end{aligned}$$

SECTION - B

81. 6

Sol. By using formula $\cos\theta \cdot \cos 2\theta \cdot \cos 4\theta \dots \cos 2^{n-1}\theta = \frac{\sin 2^n \theta}{2^n \sin \theta}$ we get

$$\log_{\sqrt{2}} \log_{1/4} \left(\frac{1}{4^2} \right)^4 = \log_{\sqrt{2}} 8 = 6$$

82. 16

Sol. If $(t_i^2, 2t_i)$, $i = 1, 2, 3$ are conormal points, then,

$$t_1 + t_2 + t_3 = 0$$

$$\text{Here } t_1 = 1, t_2 = -3 \Rightarrow t_3 = 2$$

$$\Rightarrow \alpha = 4, \beta = 4$$

83. 1

Sol. Given is equation of an ellipse with foci at (1, 2) and (-1, 2) and length of major axis being 4.

$$\Rightarrow 2ae = 2 \text{ and } 2a = 4 \Rightarrow e = \frac{1}{2}$$

84. 6

 Sol. (1, 4) is the focus of the parabola, and $3x + 4y = 4$ is the directrix

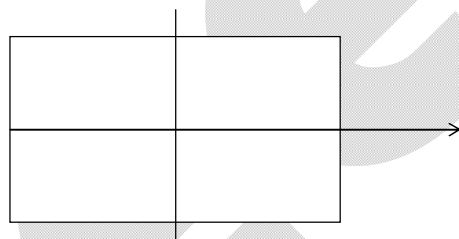
$$\Rightarrow 2a = \frac{|3 \cdot 1 + 4 \cdot 4 - 4|}{5} = 3 \Rightarrow 4a = 6$$

85. 5

 Sol. Let origin is shifter to (1, 0) after this the equation will be $\max\{|x|, |y|\} = k$

$$\text{Area} = 4k^2 = 100$$

$$\Rightarrow k = 5$$



86. 3

 Sol. $y = f(x) = (\sin^{-1} x)^3 + (\cos^{-1} x)^3, -1 \leq x \leq 1$

$$\frac{dy}{dx} = \frac{3(\sin^{-1} x)^2 - 3(\cos^{-1} x)^2}{\sqrt{1-x^2}} = 0$$

$$\Rightarrow (\sin^{-1} x + \cos^{-1} x)(\sin^{-1} x - \cos^{-1} x) = 0$$

$$\Rightarrow \sin^{-1} x = \cos^{-1} x \Rightarrow x = \frac{1}{\sqrt{2}}$$

x	-1	$\frac{1}{\sqrt{2}}$	1
f(x)	$-\frac{\pi^3}{8} + \pi^3 = \frac{7\pi^3}{8}$	$\frac{\pi^3}{64} + \frac{\pi^3}{64} = \frac{\pi^3}{32}$	$\frac{\pi^3}{8} + 0 = \frac{\pi^3}{8}$

$$\therefore \max. f(x) = \frac{7\pi^3}{8}, \min. f(x) = \frac{\pi^3}{32}$$

$$\therefore \max f(x) + \min f(x) = \left(\frac{7}{8} + \frac{1}{32}\right)\pi^3 = \frac{29}{32}\pi^3$$

$$\therefore q - p = 32 - 29 = 3$$

87. 2

 Sol. At point of maxima $f'(x) = 0$ and $f''(x) < 0$

$$\Rightarrow f''(x) = x^2 - f^2(x) \leq 0$$

 Since the curve $x^2 - y^2 = a^2$ and $x^2 - f^2(x) \leq 0$

$$x_1^2 - y_1^2 < a^2 \Rightarrow \text{Point lies outside hyperbola}$$

Hence, 2 tangents

88. 5

Sol. $\tan 55^\circ \cdot \tan 65^\circ \cdot \tan 75^\circ$

$$= \tan 5^\circ \cdot \tan(60^\circ - 5^\circ) \cdot \tan(60^\circ + 5^\circ) \cdot \frac{\tan 75^\circ}{\tan 5^\circ}$$

$$= \frac{\tan 15^\circ \cdot \tan(90^\circ - 15^\circ)}{\tan 5^\circ} = \cot 5^\circ$$

$$\therefore x = 5$$

89. 2

Sol. $f'(x) = 3x^2 + 2x - a \sin x$

$$\text{Clearly } f'(0) = 0$$

 $\Rightarrow x = 0$ must be point of minima of $y = f'(x)$ (if sign of derivative should not change)

$$\text{i.e. } f''(0) = 0$$

$$\Rightarrow f''(x) = 6x + 2 - a \cos x$$

$$f''(0) = 2 - a = 0 \Rightarrow a = 2$$

$$f'''(x) = 6 + a \sin x \Rightarrow f'''(0) = 6 + a = 6 + 2 = 8 > 0$$

90. 1

Sol. We have $f'(x) = f(x) \Rightarrow \frac{f'(x)}{f(x)} = 1 \Rightarrow \ln(f(x)) = x + c$

$$\text{As } x = 0, f(0) = 1 \Rightarrow c = 0$$

$$\text{Now, } g(x) = e^x (x+1)^2 - e^x = e^x (x^2 + 2x)$$

$$\text{So, } \int_0^1 f(x)g(x)dx = \int_0^1 e^{2x} (x^2 + 2x)dx,$$

$$\text{Put } 2x = t, dx = \frac{1}{2}dt$$

$$= \frac{1}{2} \int_0^2 e^t \left(\frac{t^2}{4} + t \right) dt$$

$$= \left(\frac{3}{4} \right) e^2 + \frac{1}{4} = ae^2 + b$$

$$\therefore (a+b) = \frac{3}{4} + \frac{1}{4} = 1$$